Year 8 SEAL Science
Energy Practice Test – Hard

Details	
Student Name:	Teacher Name:
Date Set: ____/____/26	Due Date: ____/____/26

Curriculum Assessed

On completion of this unit the student should be able to classify energy as kinetic or potential, calculate kinetic and gravitational potential energy, explain heat transfer by conduction, convection and radiation, analyse energy transfers and transformations using efficiency and Sankey diagrams, calculate appliance running costs, and evaluate ways to reduce energy use. This practice test (hard level) assesses Chapter 7 Energy.

Task Overview

Section A – Multiple Choice: 12 questions, 1 mark each (12 marks). Answer on the separate multiple-choice answer sheet.
Section B – Short Answer: 6 scenario-based questions (28 marks). Answer in this booklet.

Conditions

- Test conditions
- Formula sheet provided on page 2
- Permitted materials: pens, pencils, calculator, ruler, sharpener, eraser, highlighters.
- Not permitted: notes of any kind, blank paper, white-out, mobile phones, smart watches or any other electronic device.
- 55 mins writing time

Marking

- 40 marks
- Marked according to teacher-made marking guide.
- Show all working in calculation questions; give every final answer with a unit. When you explain or justify, use the numbers you are given.

A+	A	B+	B	C+	C	D+	D	E+	E	NS
90-100%	80-89%	75-79%	70-74%	65-69%	60-64%	55-59%	50-54%	45-49%	40-44%	<40%

Total Marks:		Percentage:		Letter Grade:	

Formula sheet

Use these formulas and units in Sections A and B.

Quantity	Formula
Kinetic energy	KE = ½ × m × v^2 (m in kg, v in m/s, KE in J)
Gravitational potential energy	GPE = m × g × h (g = 10 N/kg, h in m, GPE in J)
Efficiency	Efficiency = useful energy out ÷ total energy in × 100%
Energy used by an appliance	Energy (Wh) = power (W) × time (h)
Units	1 kWh = 1,000 Wh · 1 kW = 1,000 W
Running cost	Cost ($) = energy used (kWh) × tariff ($ per kWh)

SECTION A – Multiple-choice questions (12 marks)

Answer **all** questions on the separate multiple-choice answer sheet by filling in the letter of the best response. Each question is worth 1 mark. No marks are deducted for incorrect answers. Do **not** write your Section A answers in this booklet.

Question 1

A pole vaulter plants her pole in the box and it bends. At the moment the pole is bent the most, which type of energy is stored in the pole itself?

A. kinetic energy

B. gravitational potential energy

C. chemical potential energy

D. elastic potential energy

Question 2

A drop-tower car with its riders has a total mass of 1,500 kg. At the top of the tower it has 540,000 J more gravitational potential energy than at the bottom. How far above the bottom is the top of the tower?

A. 0.028 m

B. 36 m

C. 72 m

D. 360 m

Question 3

A roller coaster train speeds up from 10 m/s to 30 m/s as it rolls down a slope. Its mass does not change. How many times bigger is its kinetic energy at the bottom of the slope?

A. 3 times

B. 6 times

C. 9 times

D. 20 times

Question 4

A stadium is choosing panels for its new scoreboard. The table shows what each panel gives out every second when it is switched on. Which panel is the most efficient?

Panel	Light energy out each second (J)	Thermal energy out each second (J)
W	400	600
X	600	1,400
Y	150	250
Z	250	1,000

A. Panel W

B. Panel X

C. Panel Y

D. Panel Z

Question 5

Infrared heaters are fixed high above the stands at an ice rink and point down at the seats. Spectators feel warm on their skin straight away, while the air around them stays cold. How does the heat mainly reach the spectators?

A. conduction

B. convection

C. radiation

D. insulation

Question 6

At an ice rink, a layer of very cold air sits just above the ice. It stays there instead of mixing up into the stands. Why?

A. The cold air is denser than the warmer air above it, so it sinks and stays low.

B. Air particles shrink when they cool, so they sink between the warm particles.

C. The ice gives out cold, which pulls the air down and holds it there.

D. Heat can only travel upwards, so warmth never moves down to the ice.

Question 7

A stadium's video scoreboard uses 15,000 W of power. On a game night it runs for 4 hours. Electricity costs 25 cents per kWh. How much does it cost to run the scoreboard for one game night?

A. $3.75

B. $15.00

C. $1,500

D. $15,000

Question 8

Many indoor ice rinks have a shiny, foil-faced lining under the roof, above the ice. What is the main purpose of this lining?

A. It reflects cold from the ice back down onto the ice.

B. It stops convection currents from forming in the arena air.

C. It conducts heat away from the ice faster than a dull lining would.

D. It reduces the heat radiated from the warm roof onto the ice.

Question 9

In the fire-plough method, a hard stick is rubbed quickly up and down a groove in a softer piece of wood. Hot dust forms an ember, which is tipped into a tinder bundle and bursts into flame. Where does most of the energy for the flame's heat and light come from?

A. the kinetic energy of the moving stick

B. the thermal energy stored in the groove of the softer wood

C. the chemical energy stored in the tinder and wood

D. the thermal energy of the small glowing ember

Question 10

A sport scientist tests whether a new start position lowers a sprinter's 20 m time. She times three runs with each start, using a hand-held stopwatch. The three times for each start differ by up to 0.2 s. Which change would most improve the reliability of her results (how repeatable they are)?

A. Use a stopwatch that reads to 0.001 s instead of 0.01 s.

B. Record only the fastest of the three runs for each start.

C. Use a different sprinter for each type of start.

D. Do more runs with each start, such as eight, and average them.

Question 11

An energy audit of an indoor sports centre suggests four changes. Which one is a behaviour change, rather than an appliance change or a building change?

A. Staff turn the court lights down to half during training sessions.

B. The centre replaces its old floodlights with LED floodlights.

C. Builders add insulation batts to the roof of the centre.

D. The centre fits a new, more efficient heating system.

Question 12

A gymnast bounces higher and higher on a trampoline. She says: "Each bounce, the trampoline gives me more energy than I gave it." Which response is best?

A. True: the springs make new energy each time she lands on the mat.

B. False: the mat only returns energy she gives it; her leg muscles add the extra.

C. True: the mat stores up a little more energy each bounce and keeps it for the next.

D. False: nobody can ever bounce higher than the height they started from.

END OF SECTION A

SECTION B – Short-answer questions (28 marks)

Answer **all** questions in the spaces provided. Show working for all calculations. Give every final answer with a unit.

Question 1 (6 marks)

Launch coaster. A roller coaster train with its riders has a total mass of 4,000 kg. Motors launch it from rest along a flat track to a speed of 30 m/s. The train then rolls, without motors, up a hill whose top is 40 m above the launch track. Use g = 10 N/kg.

a) Assume no energy is wasted. Calculate the speed of the train as it passes over the top of the hill. Show all your working.

4 marks

b) A designer says: "With the same launch, we could make the hill 50 m tall and the train would still get over the top." Evaluate this claim. Use a calculation in your answer.

2 marks

Question 2 (5 marks)

Ice rink. To make an ice rink, very cold salty water (brine) at −9 °C is pumped through pipes set in a concrete floor. Water sprayed onto the concrete freezes into a layer of ice. A thick layer of foam insulation sits under the pipes.

a) Identify the method of heat transfer through the concrete floor.

1 mark

b) A student says: "The pipes send cold up through the concrete and into the water." Explain, in terms of particles, what really happens as the water freezes.

2 marks

c) Warm air moving in from the stands and the foyer also heats the ice. Suggest two changes to the building that would reduce this heat gain. For each change, name the method of heat transfer it reduces.

2 marks

Question 3 (4 marks)

Water slide pump. An electric pump lifts water to the top of a water slide, 12 m above the pool. Figure 1 is a Sankey diagram for the motor and pump over one second. Arrow widths are drawn to scale.

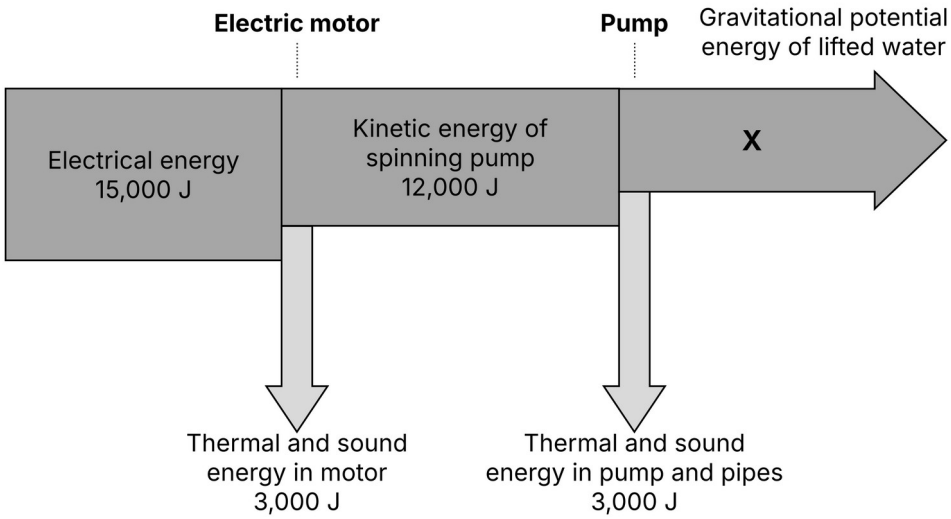


Figure 1 Energy flow through the water slide pump in one second.

a) Use Figure 1 to find the value of X, and calculate the overall efficiency of the motor and pump together.

2 marks

b) The manager wants to replace the motor with a new one that is 95% efficient. The pump will not change, and the new motor still takes in 15,000 J of electrical energy each second. She says: "Then the pump will lift 150 kg of water to the top of the slide every second." Evaluate her claim. Use calculations.

2 marks

Question 4 (5 marks)

Stadium floodlights. A football stadium has 80 floodlights. Including matches, training and community games, they are switched on for 1,200 hours a year. The current lamps are new. Electricity costs 25 cents per kWh. The club is thinking about replacing them with LED floodlights that give the same amount of light.

	Current floodlights	LED floodlights
Power of each light (W)	2,000	1,100
Lamp life (hours)	6,000	50,000
Cost to replace one lamp ($)	450	not needed within 40 years
Cost to buy and install all 80 lights ($)	already installed	130,000

a) Calculate the energy the current floodlights use in one year, in kWh.

2 marks

b) Calculate how many years the LED floodlights take to pay for themselves through lower electricity bills.

2 marks

c) The payback in part b) uses electricity costs only. Use the table to identify one other saving the LED floodlights give, and state how it changes the real payback time.

1 mark

Question 5 (2 marks)

Fire saw. At a theme park's First Nations cultural program, a ranger shows visitors the fire saw. She saws the sharp edge of one piece of wood quickly back and forth across a groove in a second, softer piece. Fine wood dust collects in the groove. After about a minute of non-stop sawing, the dust smokes and forms a glowing ember.

Most visitors who try it stop for a rest every 15 seconds. They never make an ember. Explain why, using ideas about energy.

2 marks

Question 6 (6 marks)

Pull-back toy car. In a science class, Priya investigates how far a pull-back toy car travels. Pulling the car backwards winds up a spring inside it, one click at a time. She winds the car back by 1 to 5 clicks, releases it on the smooth gym floor, and measures how far it travels. She does three trials for each number of clicks.

Clicks wound	Trial 1 (m)	Trial 2 (m)	Trial 3 (m)	Average (m)
1	0.42	0.38	0.40	0.40
2	1.05	1.13	1.09	1.09
3	2.02	1.94	1.98	1.98
4	3.12	2.05	3.20	
5	4.55	4.70	4.61	4.62

a) One result is an outlier. Identify it, and calculate the average distance for 4 clicks.

2 marks

b) Describe the relationship between the number of clicks and the distance. Quote data from the table.

2 marks

c) For some trials the gym was busy, so Priya ran the car on the rougher outdoor basketball court instead. Explain how this could affect her results, and suggest how she could fix it.

2 marks

END OF SECTION B

Additional space

Clearly number any response written here.

END OF TEST